

DS-620-F1

Data Visualization

WEEK 2

Why: Task Abstraction
How: Marks & Channels

Reda MASTOURI

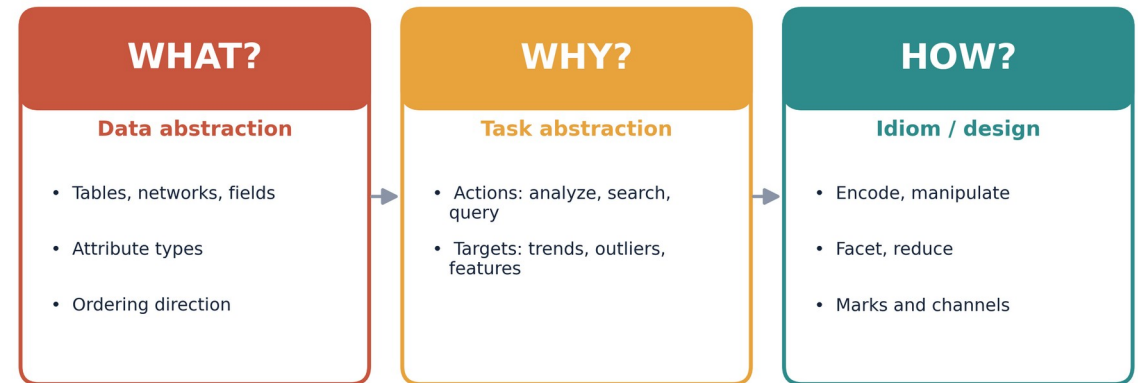
rmastouri@saintpeters.edu

READING	VAD Chapters 3 and 5
TERM	Fall 2026 Trimester
MEETS	Mondays, 6:00 – 9:25 PM
LOCATION	Dinneen Hall 326, Jersey City
FORMAT	In Person Hybrid

Week 1 ended with three questions

Every visualization answers all three. Last week we only had names for them. Tonight we fill in two.

- WHAT is shown — the data abstraction. That is Week 3.
- WHY is the user looking — the task abstraction. That is Part 1 tonight.
- HOW is it shown — the idiom, built from marks and channels. That is Part 2 tonight.
- The pairing matters: an idiom is only good relative to a task, and a task is only meaningful relative to data.
- By the end of tonight you will be able to describe any chart in this vocabulary — which is exactly what the midterm asks.



Munzner's analysis framework: every visualization answers all three questions

Munzner, Visualization Analysis and Design — the analysis framework, introduced in Chapter 1.

Two halves, three hours

1 WHY — Task abstraction

What is the user actually trying to do?

The {action, target} grammar. Analyze, search and query. Targets from trends to topology. Deriving new data rather than drawing what you were handed.

Ends with a live exercise: rate two charts against six different tasks.

2 HOW — Marks & channels

How is the picture actually built?

Marks for items and links. The channel catalogue. Expressiveness and effectiveness. Then the perception evidence: accuracy, discriminability, separability and popout.

Ends with the rules of thumb you will use every week after this.

Break around 7:35 PM, between the two halves.

Where tonight sits in the trimester

1	What's Vis, and Why Do It?	done
2	Task abstraction · Marks & channels	tonight
3	Data abstraction · Python (matplotlib, seaborn)	Assignment 1
4	Validation · Rules of thumb · Tool landscape	
5	Arrange tables · Map colour · Tableau 1	Assignment 3
6	Spatial data · Networks · Tableau 2	Project starts
7	Midterm · Project session 1	Midterm
8	Assignment review · Project session 2	
9	Tableau 3 · Project session 3	
10	Tableau 4 · Project session 4	
11	Final presentations · What's next	Project due

Week 1 asked what visualization is. Tonight asks why someone is looking — and how the picture is actually built.

The design process, in one diagram

Both halves of tonight are the two middle boxes — the translation step between a real problem and a working chart.

- Characterize the domain situation: who are the users, what is their data, what are their questions?
- Then translate, in two directions at once — the data into a data abstraction, and the domain task into an abstract task.
- Only then identify or design a suitable idiom.
- And only then worry about the algorithm that draws it efficiently.
- Skip the translation step and you end up solving a well-drawn version of the wrong problem.

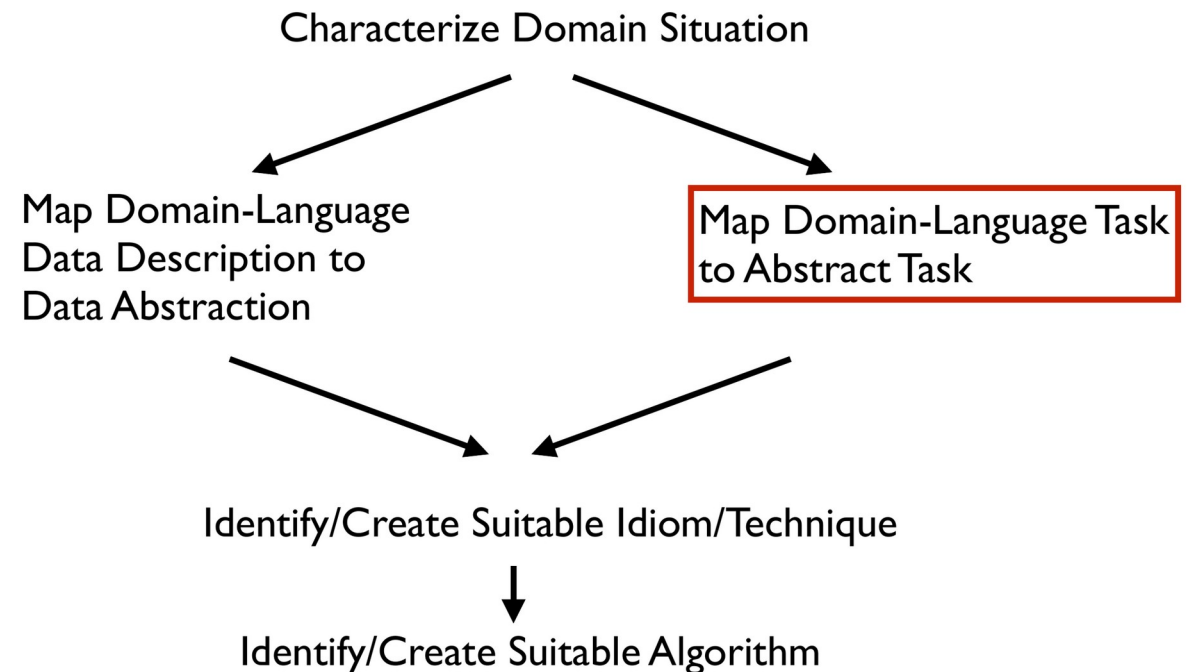


Figure from the SPU Information Visualization course deck, M. Senturk (github.com/spu-dataviz-211), after Munzner VAD.



01

PART ONE

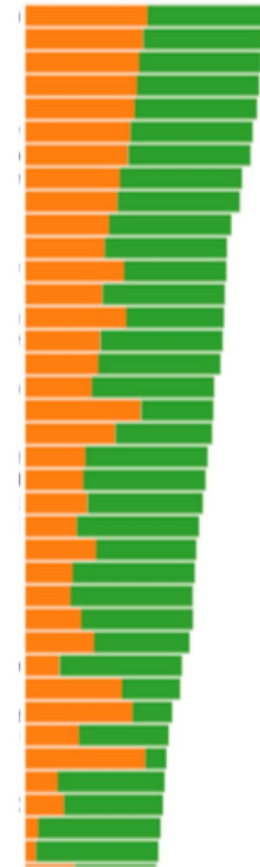
Why is the user looking at this?

The same data supports a hundred different questions, and a chart that is excellent for one of them can be useless for the next. Task abstraction is how you say which question you are serving.

Start in the domain: find me a good movie

A concrete problem, stated the way a real user would state it.

- Domain: the general public, and movie enthusiasts.
- Domain task: "what is a good movie for me?"
- Which unpacks into several different questions — highly rated by critics? by audiences? successful at the box office? similar to films I liked? in a genre I want?
- Data: is it even available? Yes — IMDB, Rotten Tomatoes and others publish ratings.
- Notice that none of this is visualization vocabulary yet. It is all domain language.



One possible idiom: a stacked bar chart of audience and critic ratings. Figure from the SPU Information Visualization course deck, M. Senturk (github.com/spu-dataviz-211), after Munzner VAD.

Then translate it into abstract vocabulary

One possible choice of data and task, restated with every domain word removed.

- Attribute: audience and critic ratings — ordinal, with three or five or ten levels.
- Attribute: genre — categorical, fewer than twenty levels.
- Items: movies — millions of them.
- Task: find high values.
- That last line is the abstraction. "Find a good movie" became "find extreme values of an ordinal attribute" — and that phrasing is what lets you reach for known idioms.

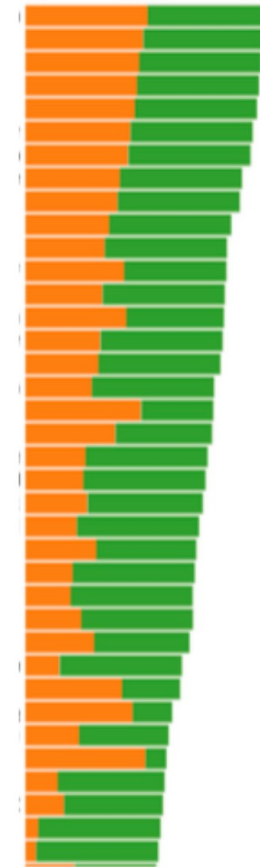


Figure from the SPU Information Visualization course deck, M. Senturk (github.com/spu-dataviz-211), after Munzner VAD.

Same task, radically different idiom

"Horrificed" plots 14,000 rated horror films from IMDB. The task is still high-scoring movies.

- Circle per item — one mark per movie.
- Circle area encodes popularity (number of votes).
- Stroke width and opacity encode average rating.
- Vertical position encodes year made.
- Interaction idiom: hovering a film draws lines to other films by the same director.



So which one is better?

The honest answer is a question back: better at what?



Horried supports browse and explore

You have time, you are in a social or casual context, you do not know exactly what you are looking for, and part of the point is enjoyment.

For that, the density and the visual pun are features, not flaws.



The stacked bar supports lookup and locate

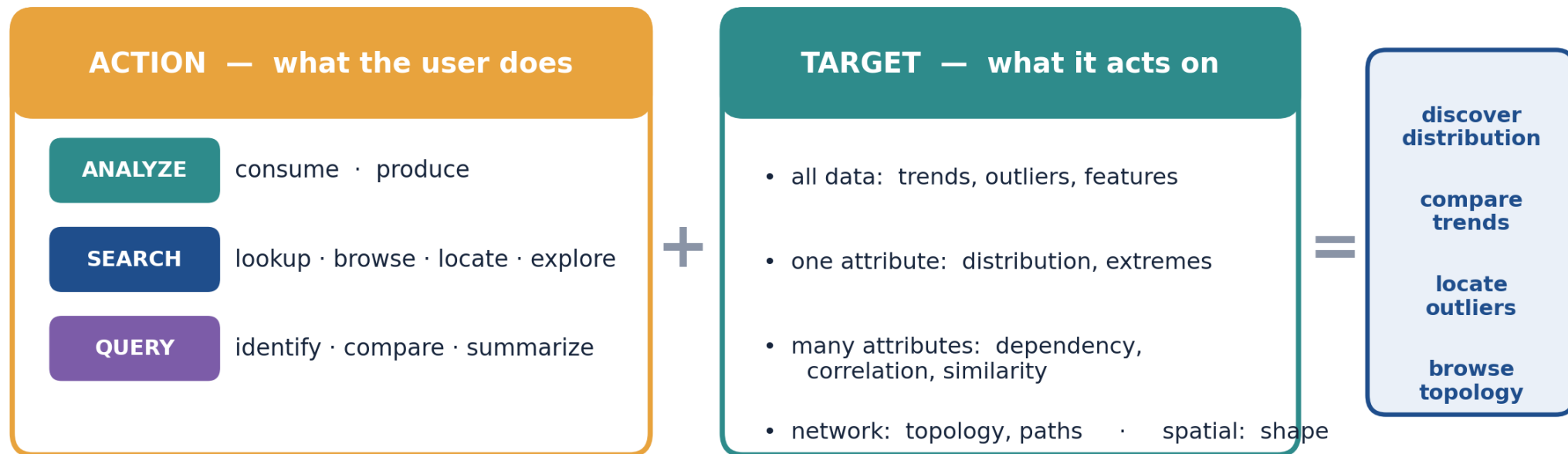
"Name the second-best-rated horror film of all time. You have ten seconds."

Now the sorted bar wins and it is not close. Sorted position on a common scale answers that in one glance.

Neither design is wrong. They serve different tasks — and if you cannot name the task, you cannot defend the design.

An abstract task is an action plus a target

Two halves: a verb describing what the user does, and a noun describing what they do it to.



Strip every word of domain jargon. “Find which genes drive the difference” becomes “identify outliers among attributes.”

Those three action levels are independent — you pick one from each, and mix and match.

After Brehmer & Munzner, A Multi-Level Typology of Abstract Visualization Tasks, IEEE TVCG 19(12), 2013; Munzner VAD Ch. 3.

Analyze: are you consuming, or producing?

The highest-level choice, and the one that changes the design most.

- CONSUME — discover: you are looking for something you don't yet know. Exploratory.
- CONSUME — present: you already know the answer and you are communicating it. Explanatory.
- CONSUME — enjoy: a casual or social reader, no analysis goal at all.
- PRODUCE — annotate: add commentary to existing data.
- PRODUCE — record: save a finding, a state, or a history.
- PRODUCE — derive: compute new data from the data you have. A crucial design choice.

➞ Analyze

➞ Consume

➞ Discover



➞ Present



➞ Enjoy



➞ Produce

➞ Annotate



➞ Record



➞ Derive

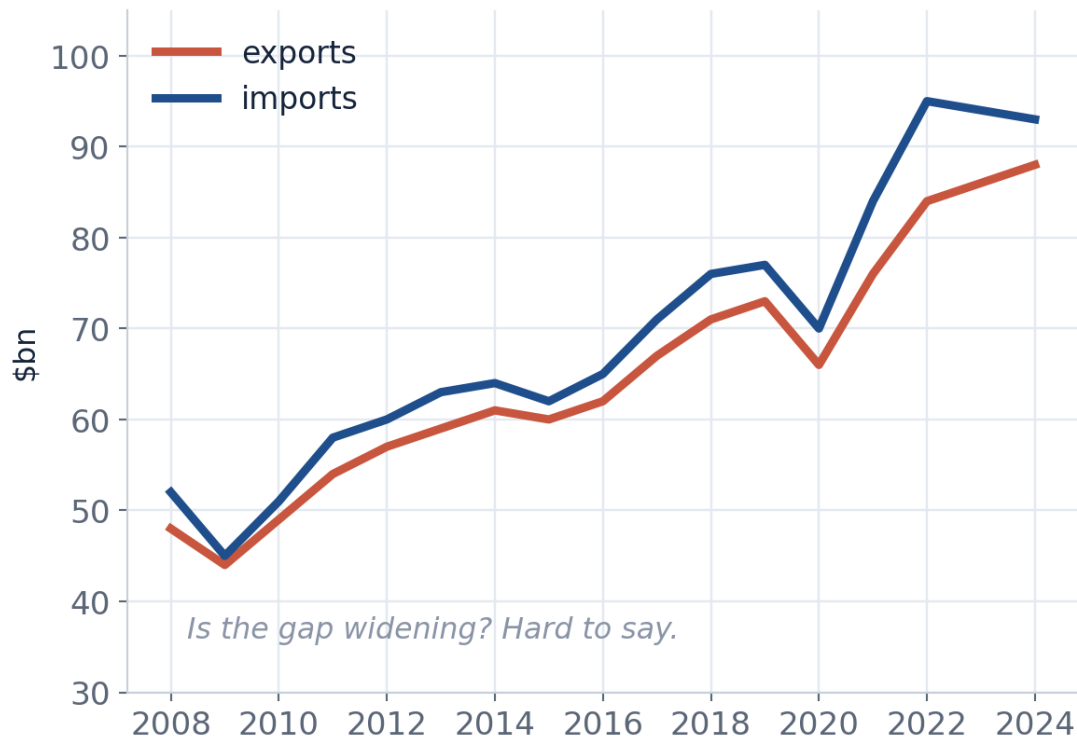


Munzner, *Visualization Analysis and Design*, Ch. 3 — figures via the SPU Information Visualization course deck (M. Senturk), after Brehmer & Munzner, *IEEE TVCG* 19(12), 2013.

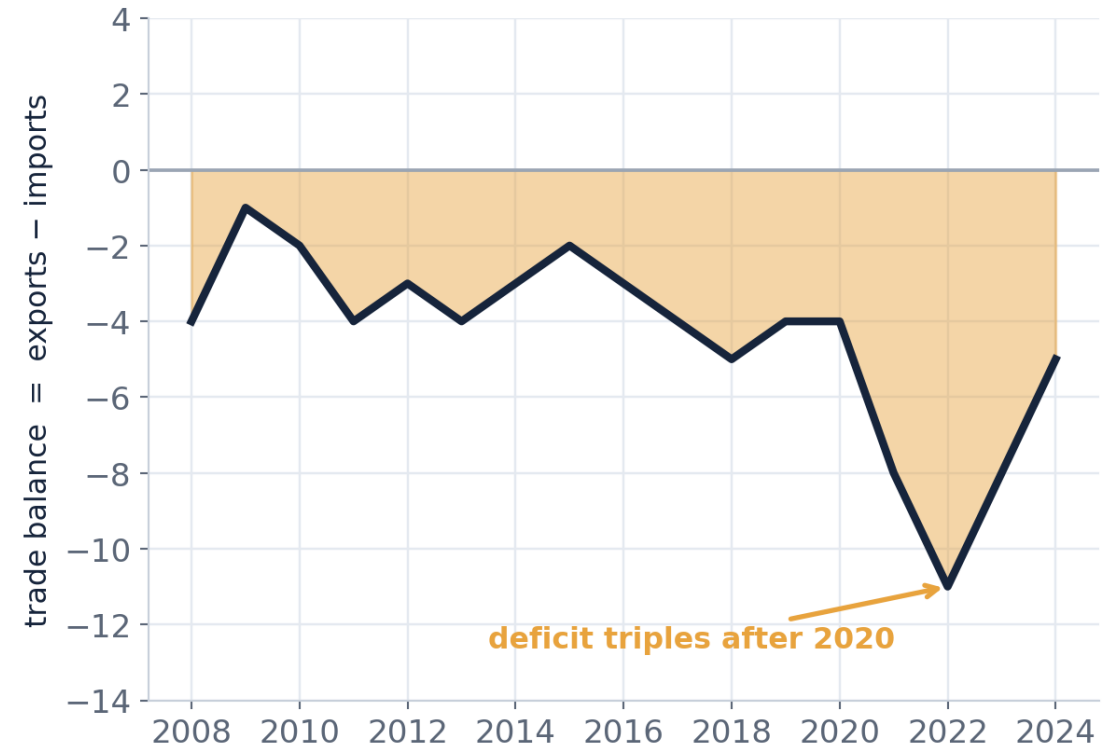
Derive: don't just draw what you were given

Decide what the right thing to show is. Create it with a series of transformations. Then draw that.

Original data: two series that track each other



Derived data: the answer, drawn directly

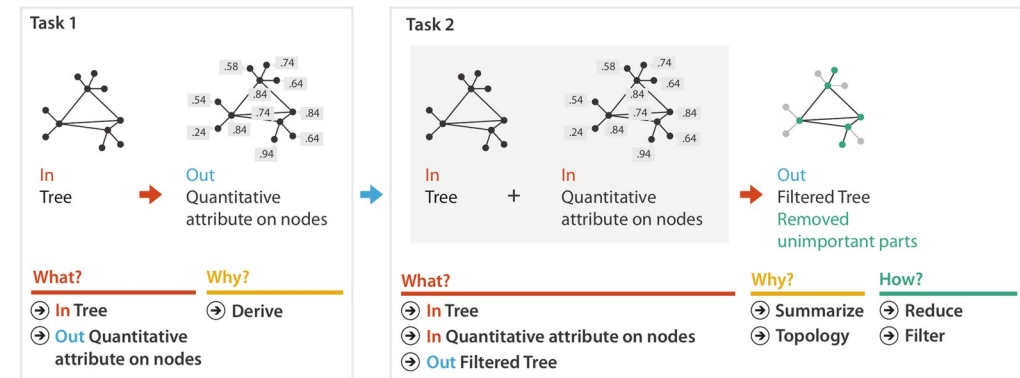


After Munzner VAD Ch. 3 — deriving a trade balance from exports and imports. Illustrative data.

Derive in practice: taming a 500,000-node tree

A real design that hinges entirely on one derived attribute.

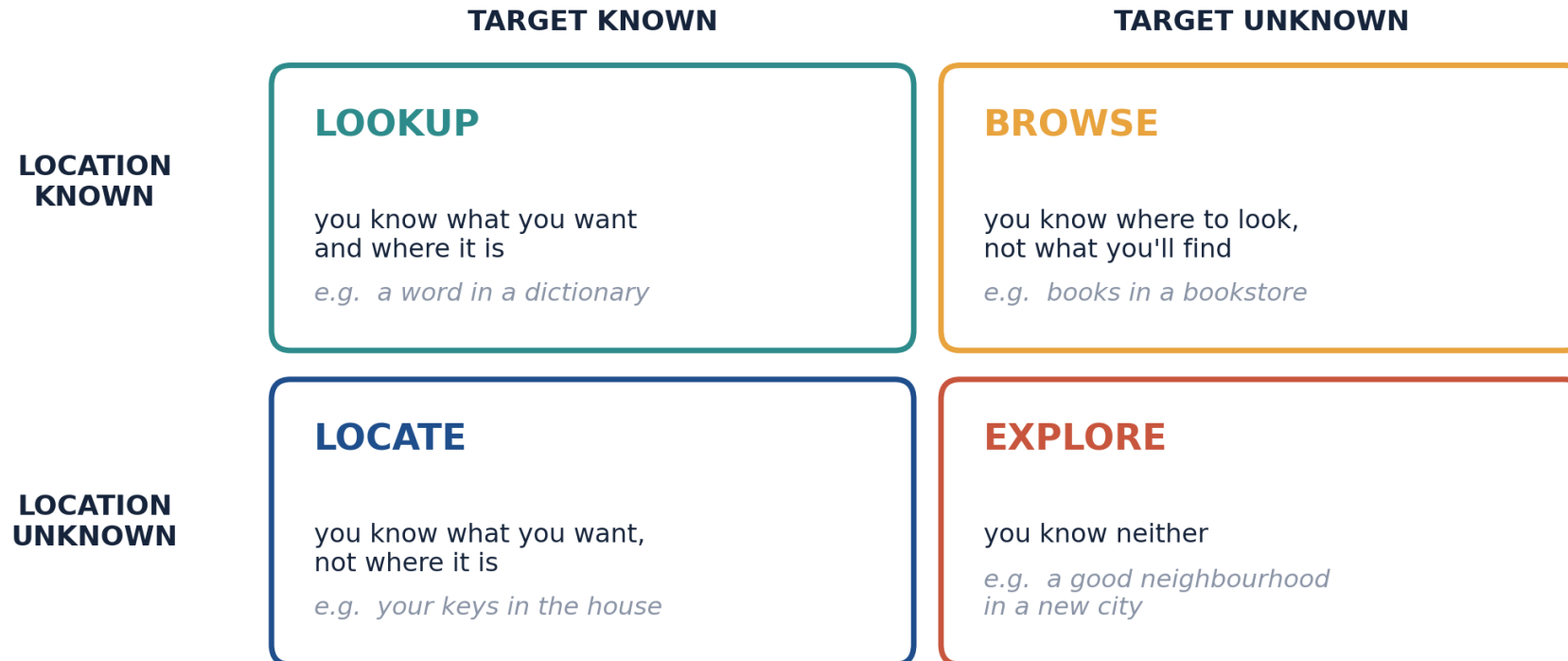
- The Strahler number is a centrality metric for trees and networks.
- It is not in the data. It is computed from the topology — a derived quantitative attribute on each node.
- Once you have it, draw only the top 5,000 nodes of 500,000 and you get a readable skeleton of the whole structure.
- Read it as two chained tasks: first derive an attribute from a tree, then use that attribute to filter the tree.
- Every non-trivial design is a chain like this, not a single task.



Auber, *Using Strahler numbers for real-time visual exploration of huge graphs*, 2002. Munzner, *Visualization Analysis and Design*, Ch. 3 — figures via the SPU Information Visualization course deck (M. Senturk), after Brehmer & Munzner, *IEEE TVCG* 19(12), 2013.

Search: it depends on what the user already knows

Two independent unknowns — do they know what they are looking for, and do they know where it is?



After Munzner VAD Ch. 3 / Brehmer & Munzner 2013.

Query: once you have found it, what do you ask?

And crossed with scope — does one item matter, some of them, or all of them?



Crossed with how much data matters: one · some · all — three independent choices you mix and match.

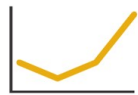
After Munzner VAD Ch. 3 / Brehmer & Munzner 2013.

Targets: the things those actions act on

Some targets apply to all data; others are specific to attributes, networks or space.

→ All Data

→ Trends



→ Outliers



→ Features



→ Attributes

→ One

→ Distribution



→ Extremes

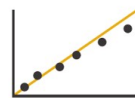


→ Many

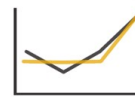
→ Dependency



→ Correlation



→ Similarity



→ Network Data

→ Topology



→ Paths



→ Spatial Data

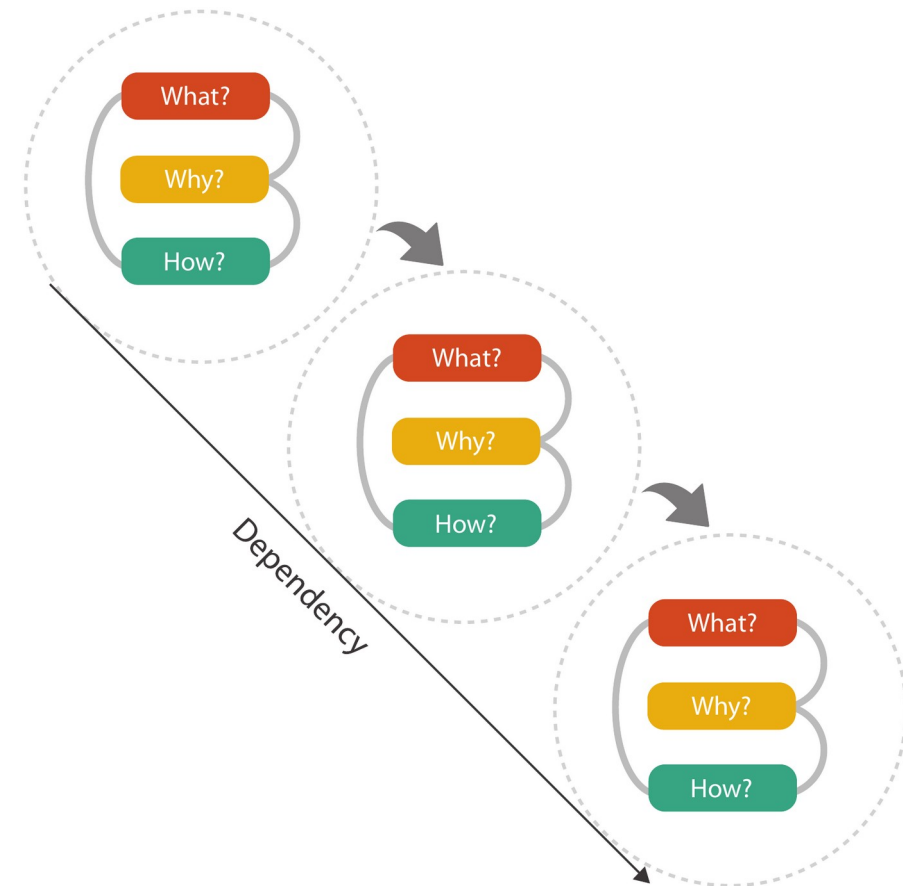
→ Shape



Real designs are chains, not single tasks

The output of one task becomes the input to the next. Munzner calls this means and ends.

- An abstract task rarely stands alone.
- What you derive at one step becomes the what of the next step.
- Why you needed it at one step becomes the how of the step above.
- The Strahler example was exactly this: derive, then filter using what you derived.
- When you analyze someone else's design, expect to find three or four linked tasks, not one.



15

Munzner, *Visualization Analysis and Design*, Ch. 3 — figures via the SPU Information Visualization course deck (M. Senturk), after Brehmer & Munzner, *IEEE TVCG* 19(12), 2013.

How to actually do this

1 Strip every domain word

Rule of thumb: systematically remove all domain jargon.

If your task statement still says "genes", "claims" or "shipments", you have not abstracted it yet.

2 Use data vocabulary for targets

You need the data abstraction inside the task abstraction, because targets are described in data terms — attributes, items, links.

The two abstractions are not independent.

3 Iterate, don't sequence

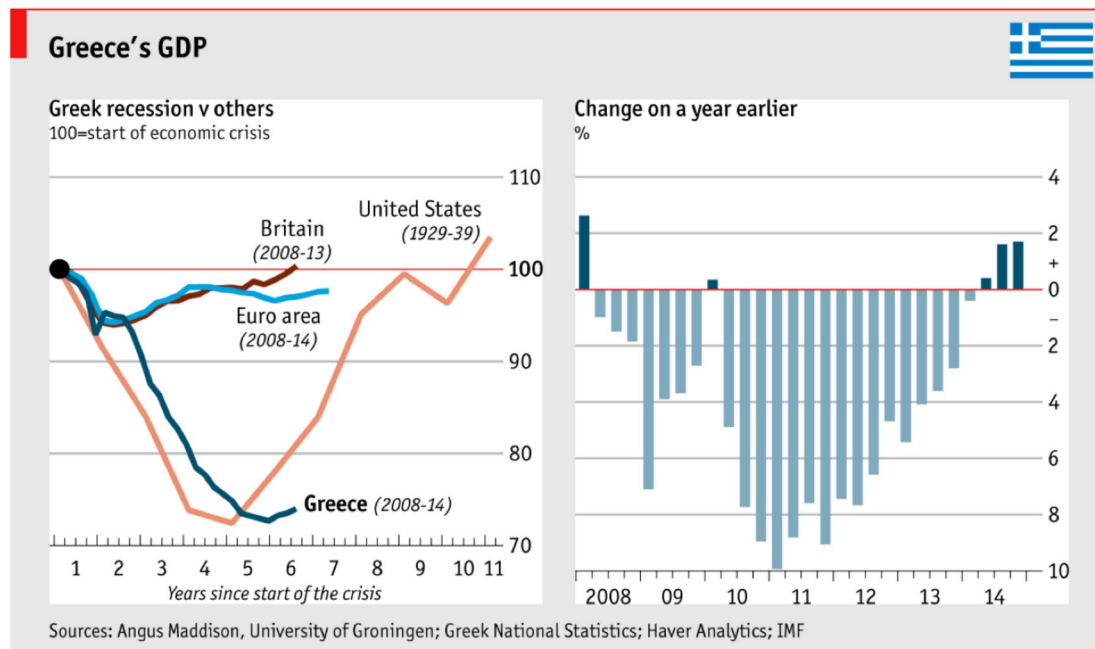
First pass at the data, first pass at the task, second pass at the data, second pass at the task.

Task abstraction often reveals that the data must be transformed — which changes the data abstraction.

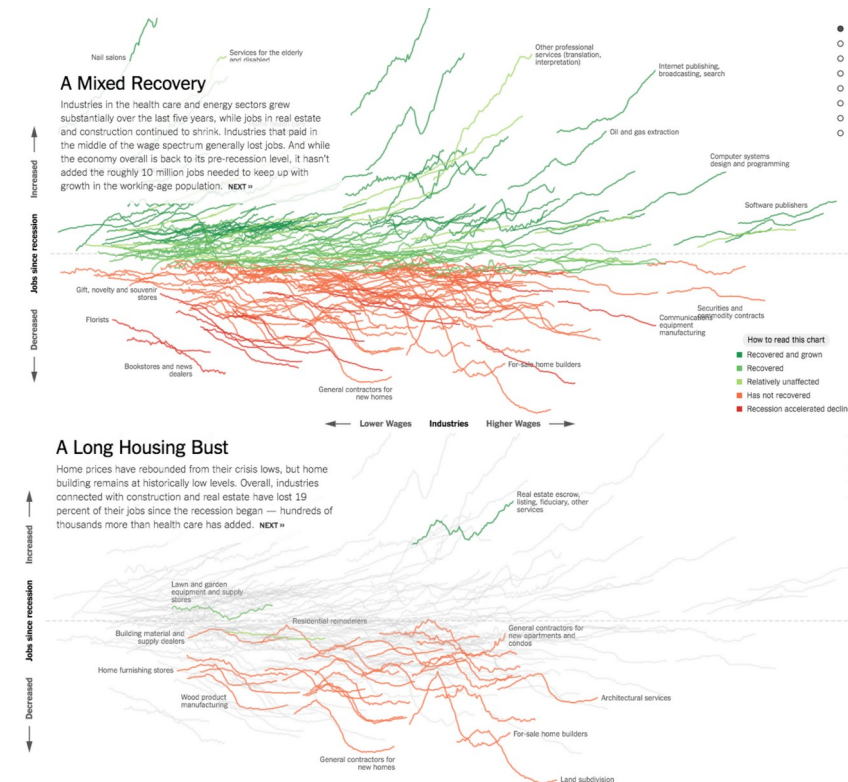
The {action, target} pairs are a good starting vocabulary — but sometimes you will need more precision than they offer, and that is fine.

Two published designs, read in this vocabulary

Left: compare and derive. Right: trends and outliers across 255 series at once.



The Economist — Greece's GDP

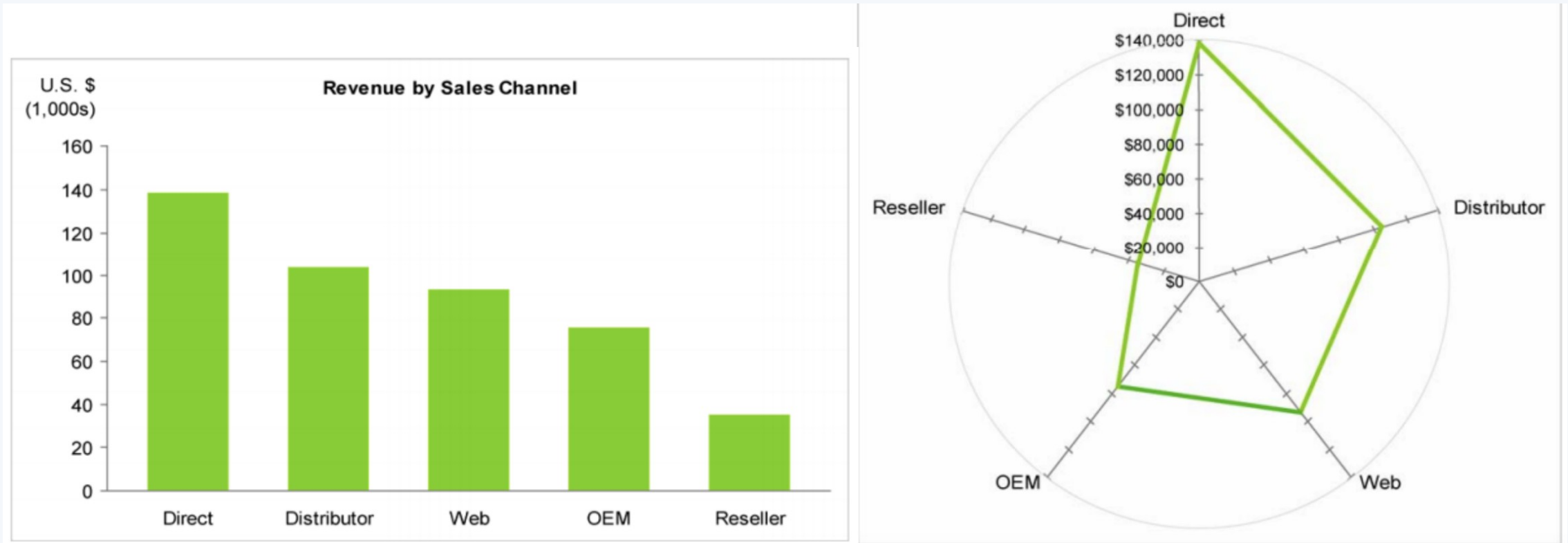


NYT Upshot — the recession in 255 charts

The Economist; NYT Upshot, [nytimes.com/interactive/2014/06/05/upshot/](https://www.nytimes.com/interactive/2014/06/05/upshot/). Munzner, *Visualization Analysis and Design*, Ch. 3 — figures via the SPU Information Visualization course deck (M. Senturk), after Brehmer & Munzner, *IEEE TVCG 19(12)*, 2013.

Exercise: rate these two charts, six times over

Same five numbers, two encodings. Score each chart out of five for each task — work alone for four minutes, then compare with your neighbour.



https://www.perceptualedge.com/articles/dmreview/radar_graphs.pdf

23

Exercise after Marti Hearst; charts from Stephen Few, [perceptualedge.com/articles/dmreview/radar_graphs.pdf](https://www.perceptualedge.com/articles/dmreview/radar_graphs.pdf). Munzner, *Visualization Analysis and Design*, Ch. 3 — figures via the SPU Information Visualization course deck (M. Senturk), after Brehmer & Munzner, *IEEE TVCG* 19(12), 2013.

What the exercise is designed to show

A Sorting and outliers

The bar chart wins easily. Length on a common baseline, already sorted.

On the radar chart you are comparing distances from a centre point at different angles — much harder.

B/C The interesting rows

Adjacent pairs on the radar chart are easy to compare; non-adjacent pairs are not.

So the same chart scores differently on two tasks that look identical on paper. Position around the ring is an arbitrary design choice with real consequences.

D/F Shape and engagement

For an overall gestalt, or for catching attention, the radar chart is defensible — the enclosed shape is memorable.

Which is why these charts persist despite everything in Part 2.

The lesson is not "radar charts are bad." It is that a chart's quality is a function of the task, and you now have language for saying which task.



02

PART TWO

How is the picture actually built?

Every chart you have ever seen is a combination of two things: geometric marks, and visual channels that control how those marks look. That is the entire vocabulary.

Before we start: two numbers

Nine and twenty-six. How many different ways can you show them?

1 Solo — three minutes

Sketch three different ways to visually represent the two numbers 9 and 26.

Pen and paper. No laptops. Rough is fine.

2 Pairs — two minutes

Compare with your neighbour.

How many of your three matched theirs?
Usually one or two — almost everyone draws bars first.

3 Together — three minutes

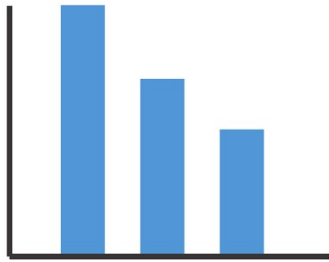
Now sketch two more that neither of you had.

This is where it gets interesting: area, angle, colour, position, count, texture, nesting.

Keep your sketches — we will name every one of them in the next ten minutes. There are at least 45 defensible answers.

Visual encoding = marks + channels

Four charts. Each one adds exactly one channel to the one before it.



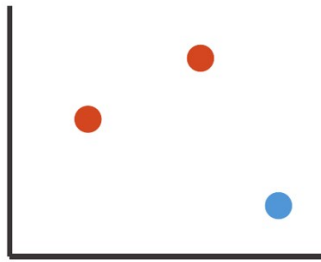
1:
vertical position

mark: line



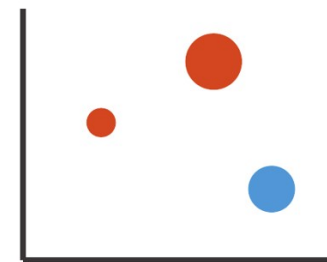
2:
vertical position
horizontal position

mark: point



3:
vertical position
horizontal position
color hue

mark: point

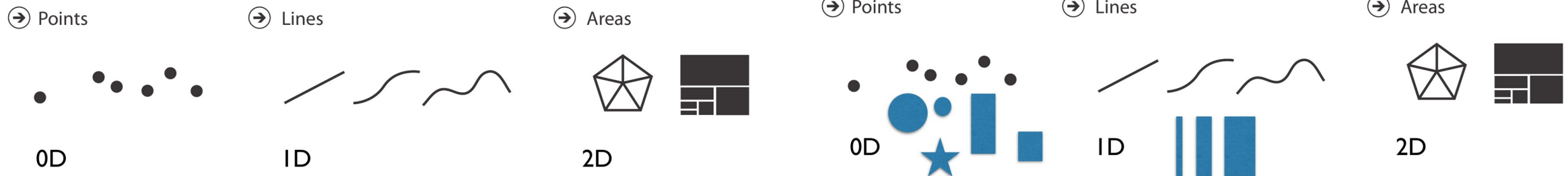


4:
vertical position
horizontal position
color hue
size (area)

mark: point

Marks for items: points, lines, areas

Classified by dimensionality — 0D, 1D, 2D. Volume marks exist but are rarely useful.



The three geometric primitives

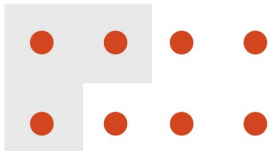
Mark type constrains what else you can encode

Munzner, *Visualization Analysis and Design*, Ch. 5 — via the SPU Information Visualization course deck (M. Senturk), after Alex Lex & Miriah Meyer, dataviscourse.net.

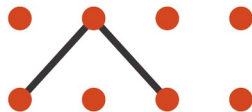
Marks for links: containment and connection

Marks do not only stand for items. They can stand for relationships between items.

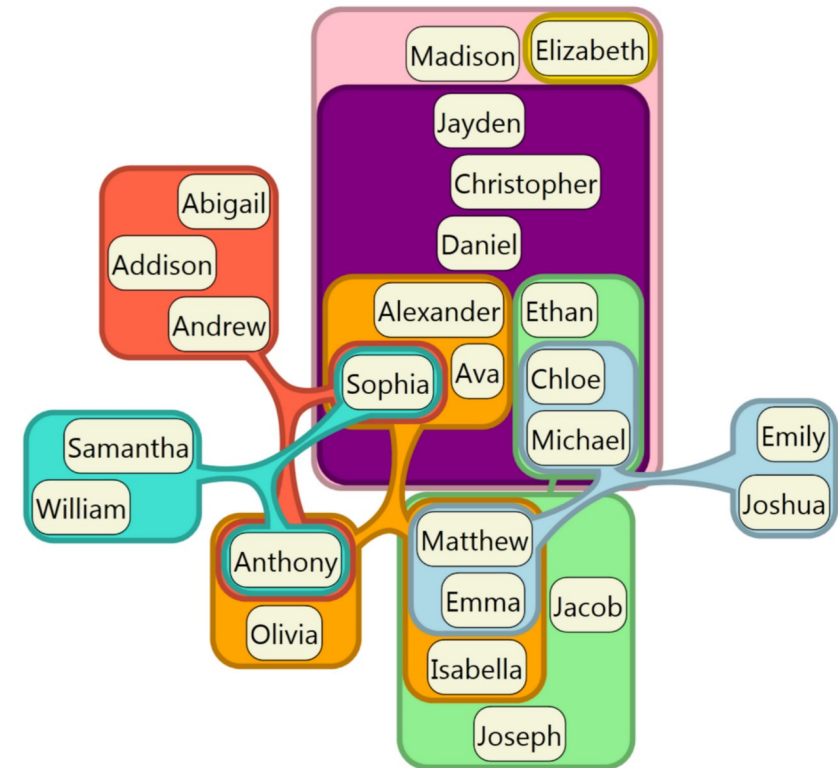
➔ Containment



➔ Connection



Two ways to show that items are related



Containment nests; connection does not

Right: Riche & Dwyer, *Untangling Euler Diagrams*, 2010. Munzner, *Visualization Analysis and Design*, Ch. 5 — via the SPU Information Visualization course deck (M. Senturk), after Alex Lex & Miriah Meyer, dataviscourse.net.

Channels: the ways you can vary a mark

Position, colour, shape, tilt, size — and each of those subdivides further.

➞ Position

➞ Horizontal



➞ Vertical



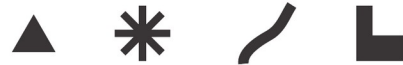
➞ Both



➞ Color



➞ Shape



➞ Tilt



➞ Size

➞ Length



➞ Area

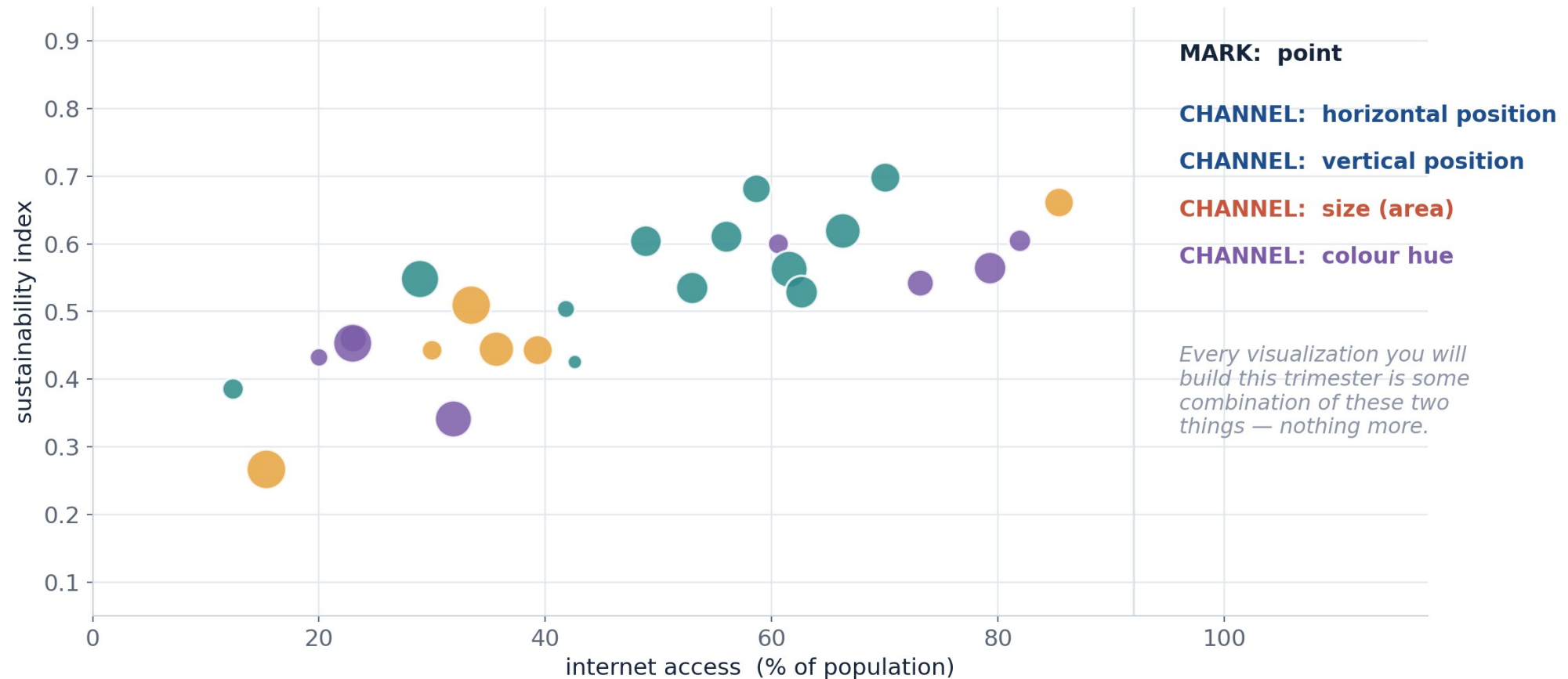


➞ Volume



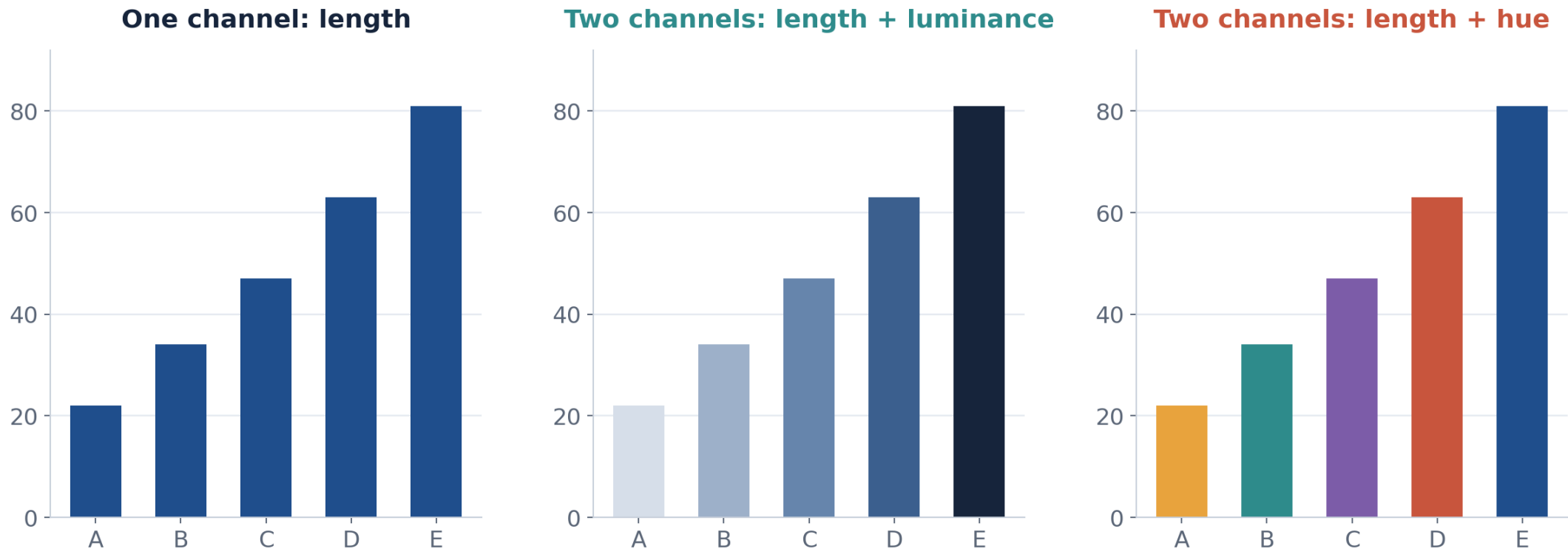
Putting it together: one chart, decomposed

Point marks, four channels, four attributes. Nothing else is going on.



Redundant encoding: same attribute, two channels

A stronger message — but you have spent a channel to send it.



Middle: luminance is ordered, so it reinforces the ranking — redundant and correct. Right: hue is unordered, so it implies five categories that do not exist.

What is wrong with this picture?

A local news graphic showing snowfall: 8 inches, 17 inches, 180 inches, 656 inches.

- Look at the bar heights. Then look at the numbers.
- 656 is 82 times 8. The bar is roughly four times as tall.
- The channel is not proportional to the data at all — the bars appear to have been sized by eye.
- Everything on the graphic is factually correct. The labels are right. The encoding is not.
- Rule: if you use a magnitude channel, it must be proportional to the value it encodes.



Via twitter.com/ChaseThomason. Munzner, *Visualization Analysis and Design*, Ch. 5 — via the SPU Information Visualization course deck (M. Senturk), after Alex Lex & Miriah Meyer, dataviscourse.net.

The two principles that decide everything

Expressiveness answers "may I use this channel?" Effectiveness answers "should I?"

EXPRESSIVENESS

Match the channel type to the data type.

Ordered data → a magnitude channel

position, length, area, luminance

Categorical data → an identity channel

spatial region, colour hue, shape, motion

Putting an unordered category on a magnitude channel invents an order that is not in the data.

EFFECTIVENESS

Not all channels are equally accurate.

Rank the channels by accuracy

Rank your attributes by importance

Match the two lists top-down

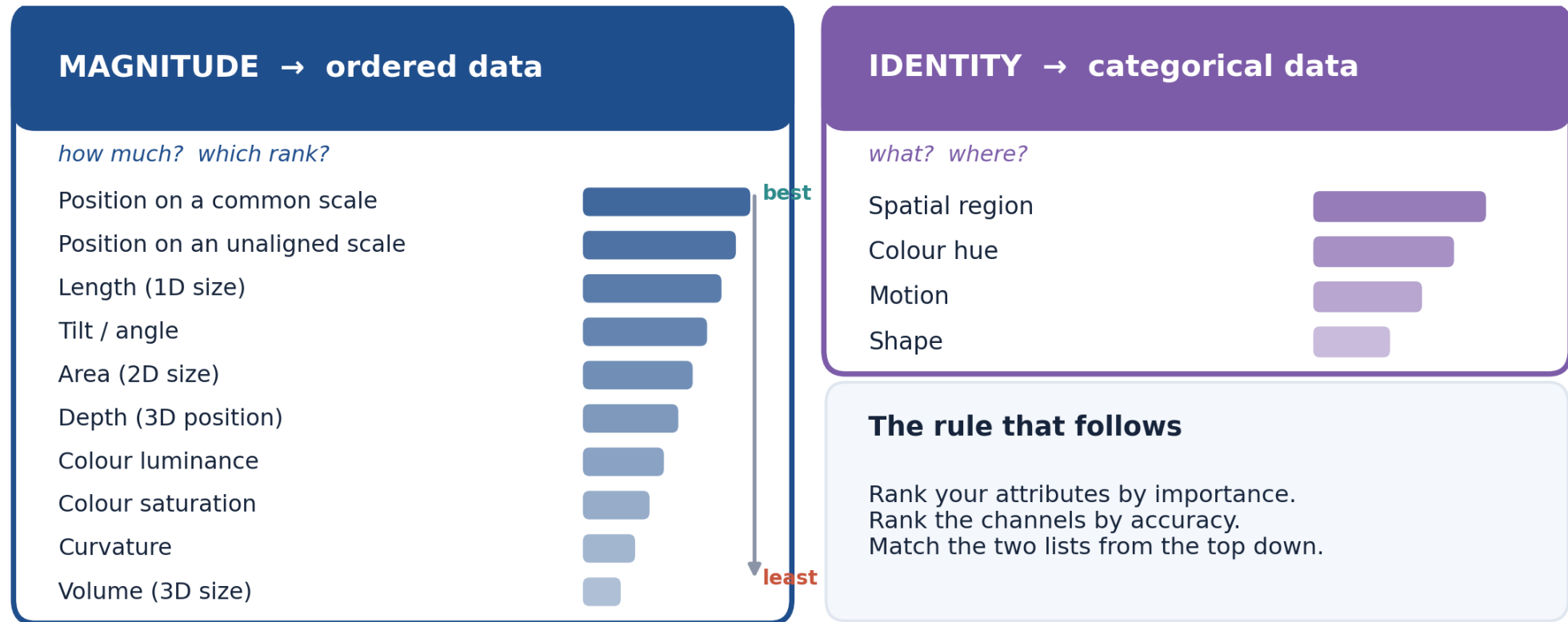
Most important attribute → best channel

Position on a common scale wins.
Area, luminance and curvature are read far less precisely.

After Munzner VAD Ch. 5; effectiveness rankings after Cleveland & McGill (1984) and later replications.

Which means channels come in two families

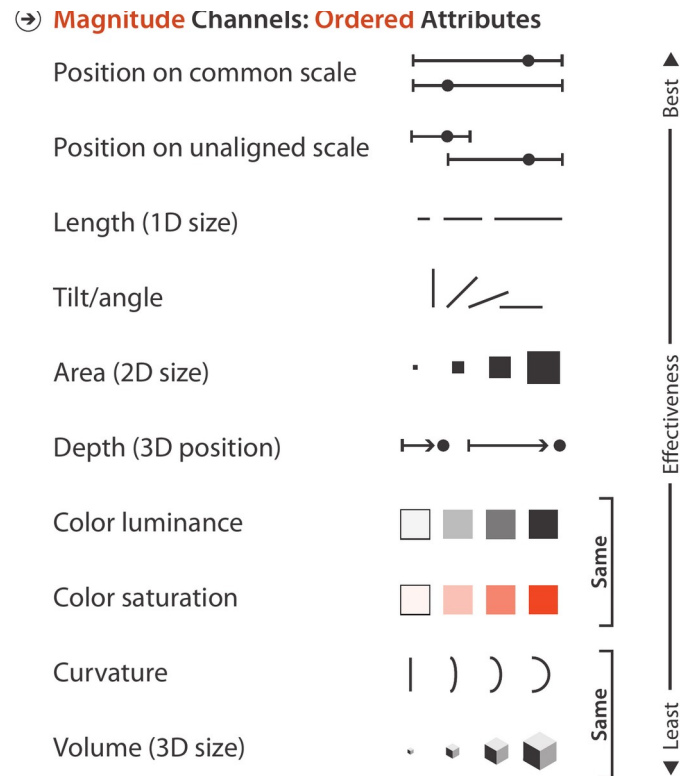
Magnitude channels for ordered data. Identity channels for categorical data. Mixing them up breaks expressiveness.



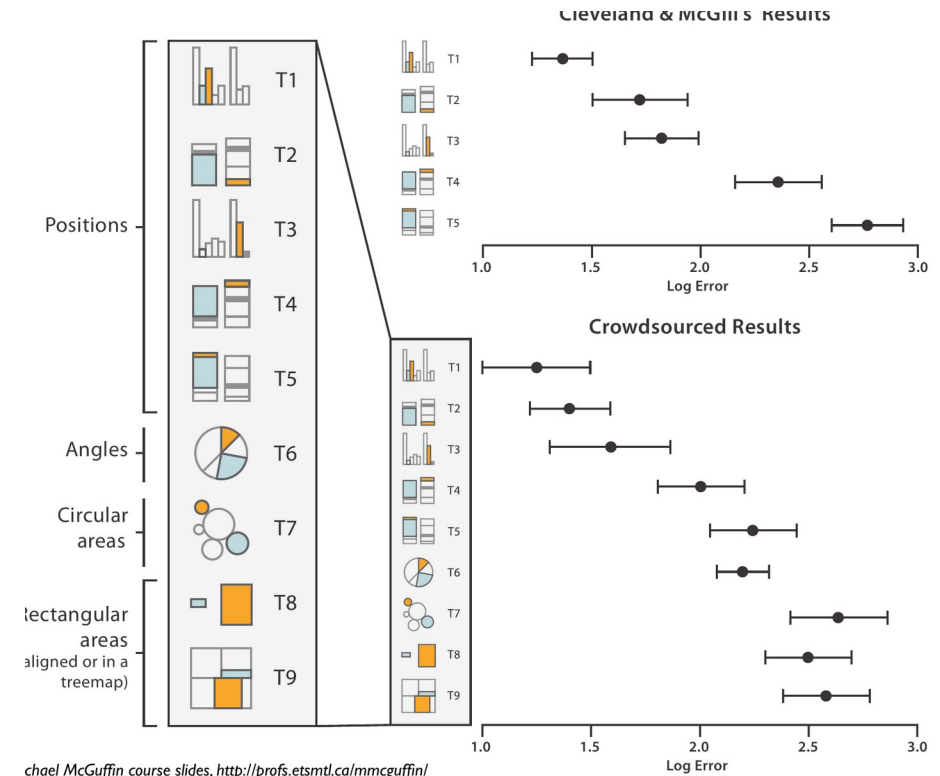
After Munzner VAD Ch. 5.

The ranking is not an opinion — it was measured

Cleveland and McGill ran the experiments in 1984. Heer and Bostock replicated them with crowdsourcing in 2010 and got the same order.



The magnitude channels, ranked by accuracy



Measured log error, two studies, same ordering

Cleveland & McGill, JASA 79:387 (1984); Heer & Bostock, CHI 2010. Munzner, Visualization Analysis and Design, Ch. 5 — via the SPU Information Visualization course deck (M. Senturk), after Alex Lex & Miriah Meyer, dataviscourse.net.

Four questions to ask about any channel

1 Accuracy

How precisely can people tell the difference between two encoded values?

Measured experimentally, and governed by a power law — some channels are systematically magnified, others compressed.

2 Discriminability

How many unique steps can people actually perceive?

If your attribute has more levels than the channel has usable steps, the encoding fails no matter how accurate it is.

3 Separability

Is the ability to read this channel affected by another channel being used at the same time?

Some pairs are fully independent; others interfere badly.

4 Popout

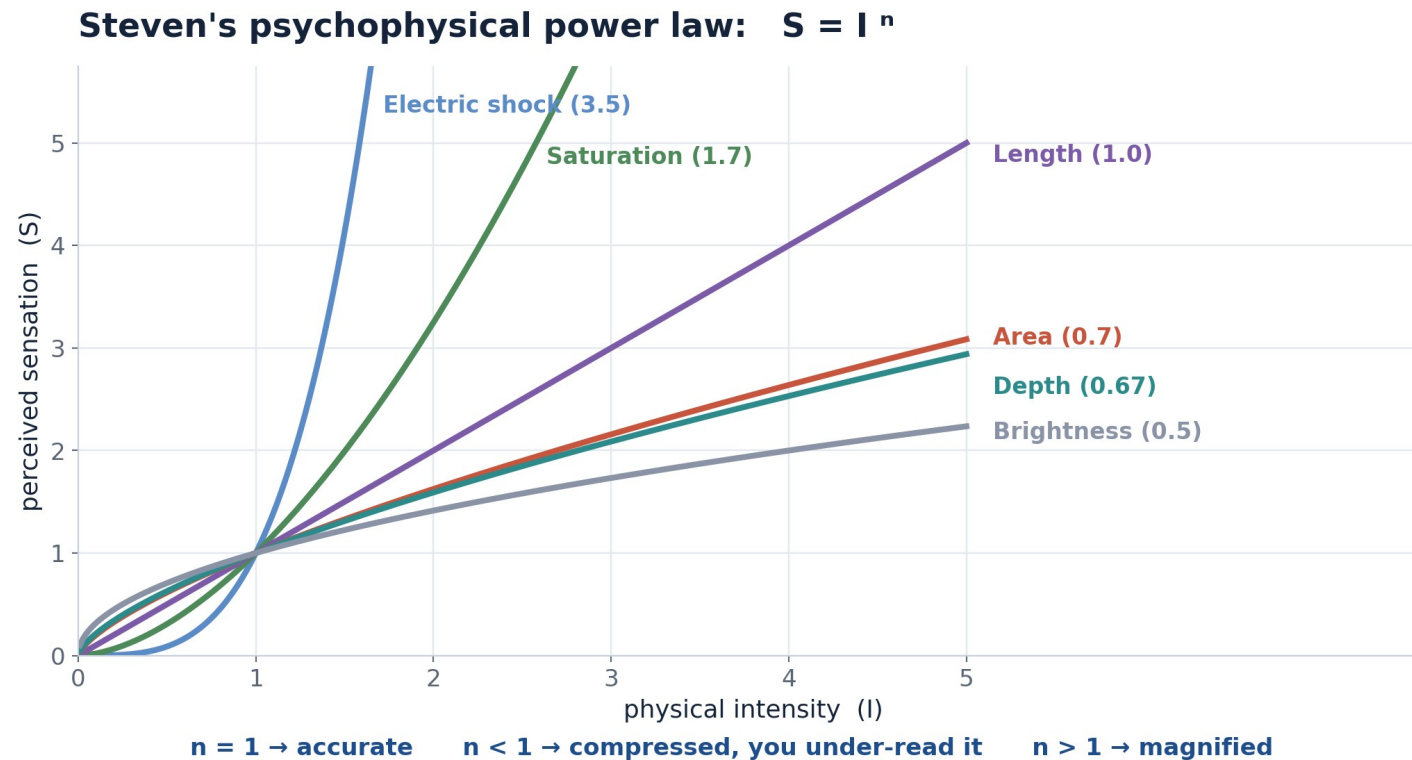
Can a value jump out without conscious search?

Some channels support parallel processing across the whole visual field. Most combinations of two channels do not.

The next four slides take these one at a time, with the evidence.

Accuracy: perception is not linear

Steven's psychophysical power law. Sensation scales as intensity raised to an exponent, and the exponent depends on the channel.

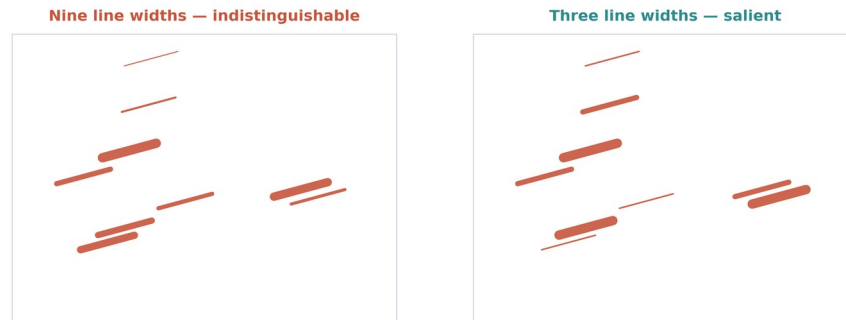


Length has an exponent of 1. Area (0.7) and brightness (0.5) are compressed — doubling them does not read as double.

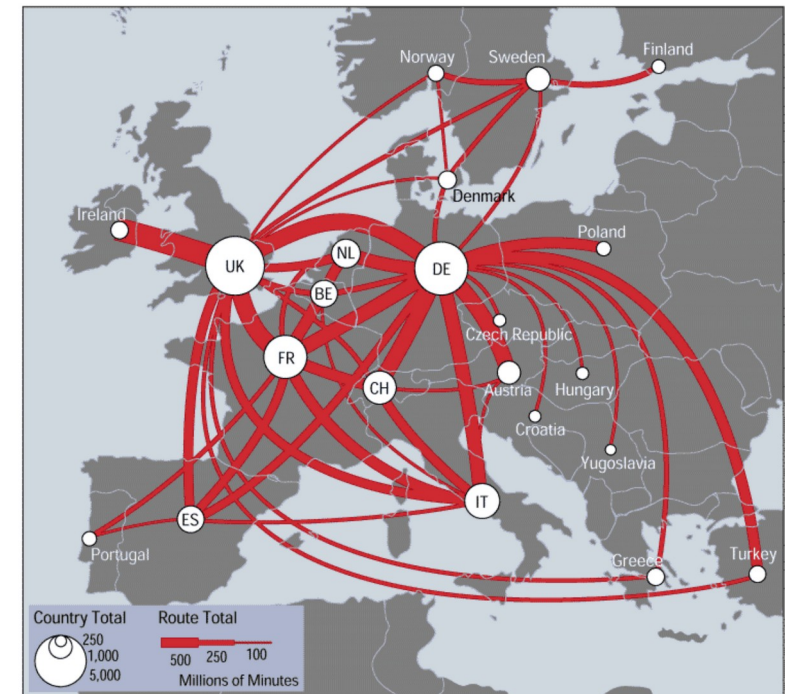
Steven's power law, $S = I^n$. After Munzner VAD Ch. 5.

Discriminability: how many steps can people actually see?

A channel with four usable bins cannot carry an attribute with twelve levels.



Line width supports only a handful of usable bins. If your attribute has more levels than the channel has steps, the encoding fails — bin the data, or pick another channel.



[mappa.mundi.net/maps/maps_014/telegeography.html]

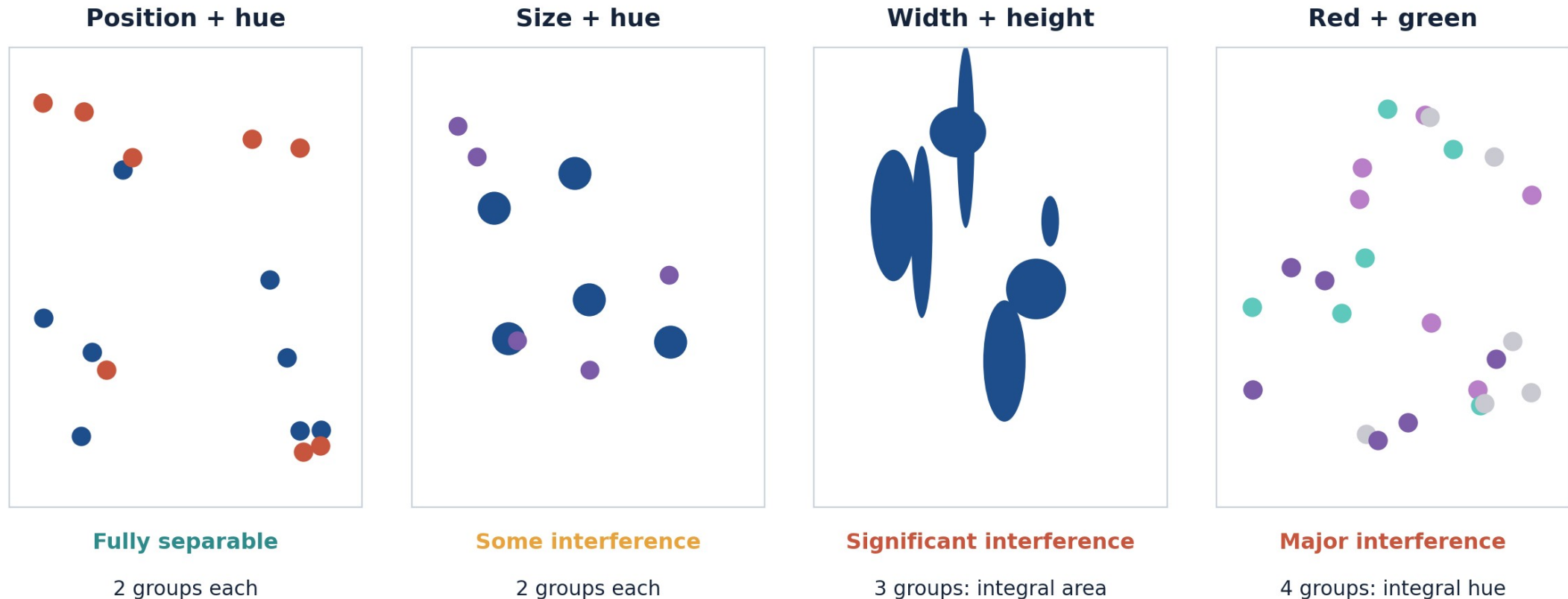
Line width supports only a few distinguishable steps

Used well: few bins, but highly salient

Right: mappa.mundi.net telegeography map. Munzner, *Visualization Analysis and Design*, Ch. 5 — via the SPU Information Visualization course deck (M. Senturk), after Alex Lex & Miriah Meyer, dataviscourse.net.

Separability: channels can interfere with each other

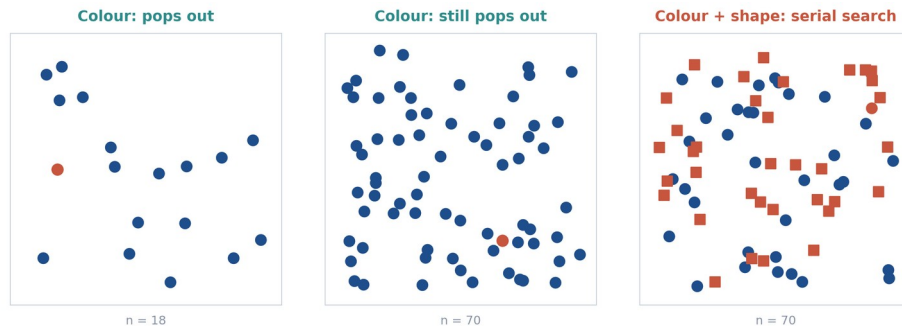
Channel pairs sit on a spectrum from fully independent to completely fused.



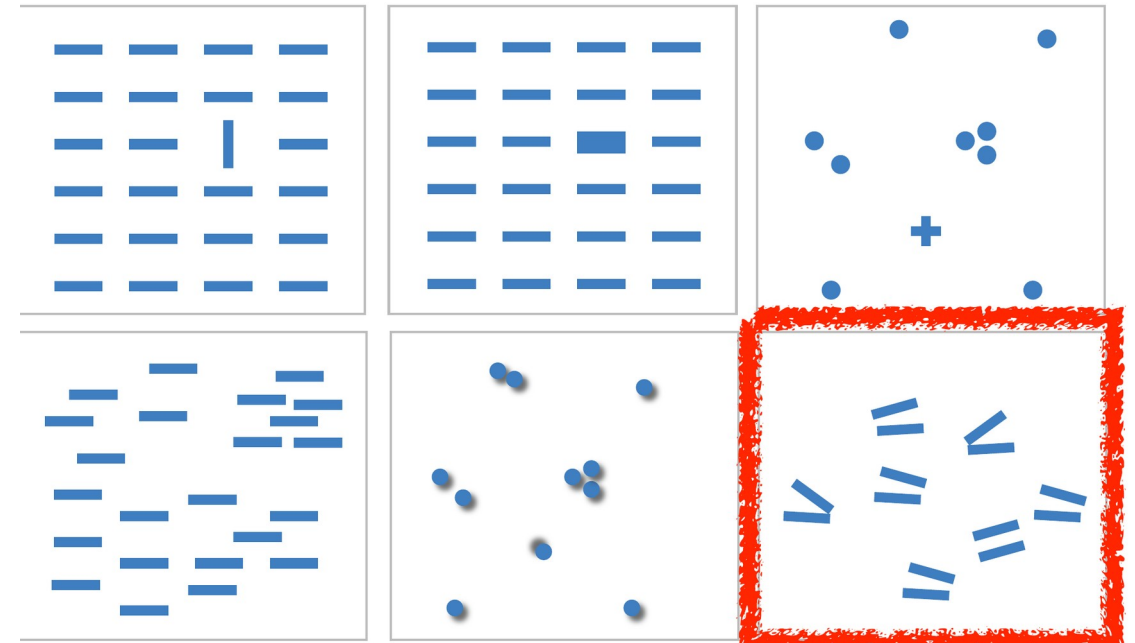
After Munzner VAD Ch. 5.

Popout: when search becomes free

Find the red circle. Then find it again with twice as many distractors.



Single channel → parallel processing: speed is independent of how many distractors there are. Conjunction of two channels → serial search: speed depends on the count.



One channel is parallel; a conjunction is serial

Many channels pop out — but not all

Munzner, *Visualization Analysis and Design*, Ch. 5 — via the SPU Information Visualization course deck (M. Senturk), after Alex Lex & Miriah Meyer, dataviscourse.net.

Grouping: four ways to say "these belong together"

Two use link marks; two use identity channels.

Marks as Links

➔ Containment



➔ Connection



➔ Identity Channels: Categorical Attributes

Spatial region



Color hue



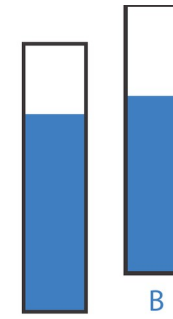
Motion



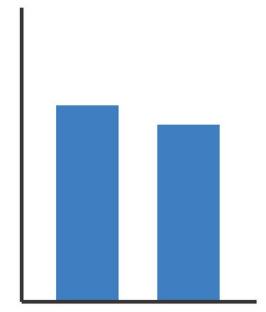
Shape



A B
length



A B
position along
unaligned
common scale



A B
position along
aligned scale

40

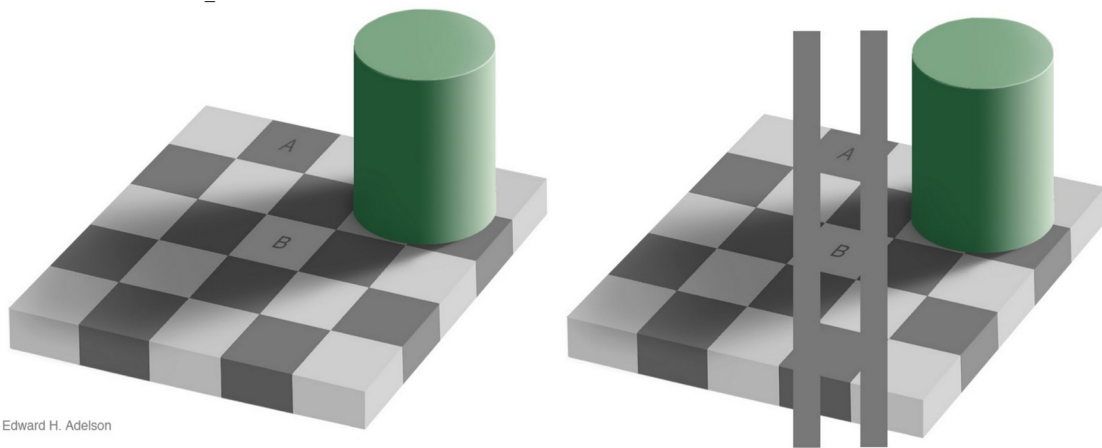
Containment, connection, proximity, similarity

And why a common frame beats a bare comparison

Munzner, *Visualization Analysis and Design*, Ch. 5 — via the SPU Information Visualization course deck (M. Senturk), after Alex Lex & Miriah Meyer, dataviscourse.net.

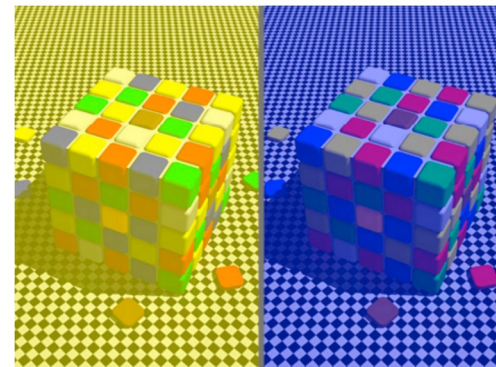
Nothing is judged in isolation

Luminance and colour are read relative to their surroundings — which is a problem if you are encoding data with them.



Edward H. Adelson

Squares A and B are the identical grey



The tiles read as orange and purple. They are the same.

Left: Adelson checkershadow, persci.mit.edu/gallery/checkershadow. Right: purveslab.net/seeforyourself. Munzner, *Visualization Analysis and Design*, Ch. 5 — via the SPU Information Visualization course deck (M. Senturk), after Alex Lex & Miriah Meyer, dataviscourse.net.

The rules of thumb you now have

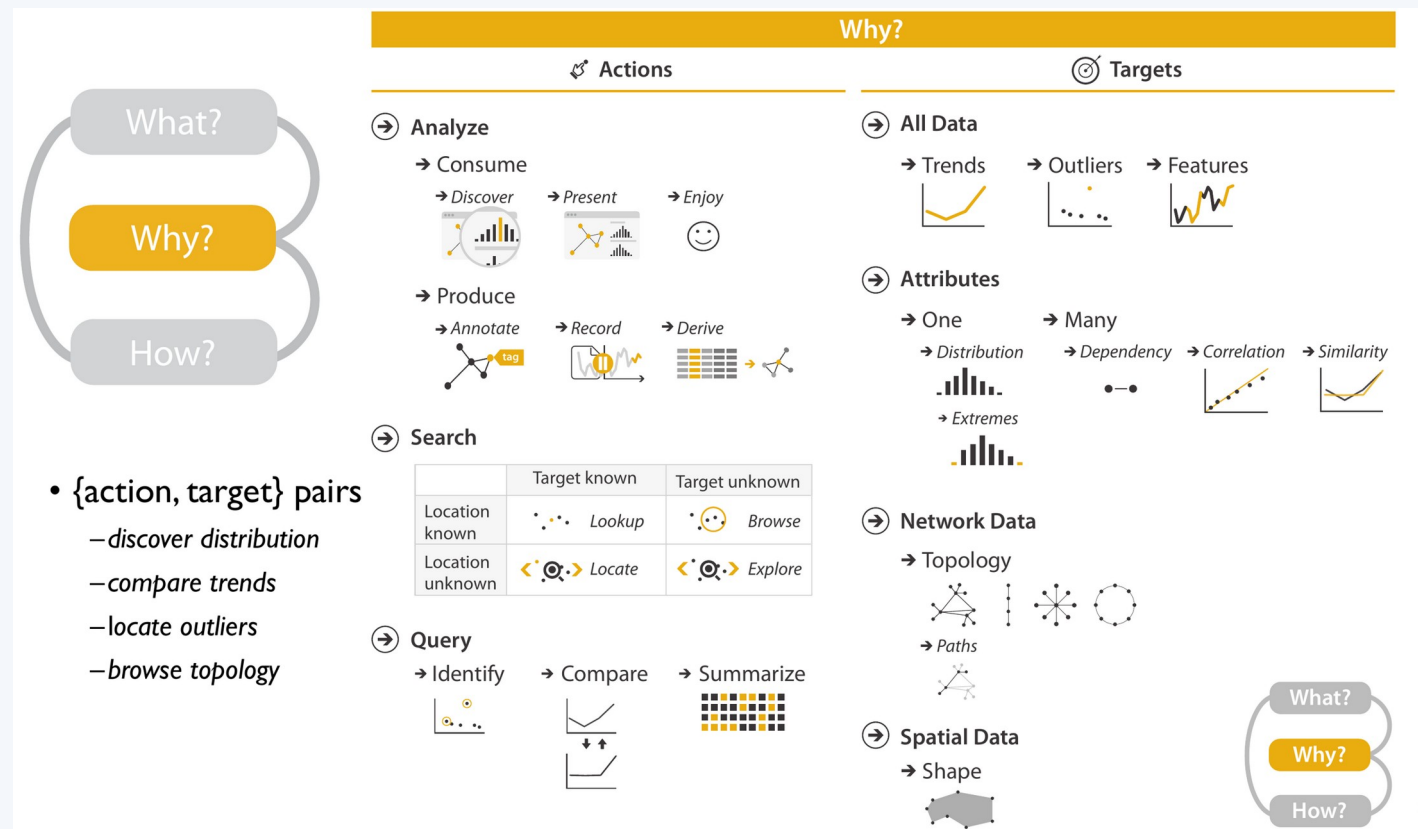
Everything in tonight's second half collapses into a handful of usable rules.

- 1 Name the task before you judge the chart. Action plus target, no domain jargon.
- 2 Ask whether you should derive a new attribute instead of drawing the raw columns.
- 3 Match channel type to data type: magnitude for ordered, identity for categorical.
- 4 Spend position — the most accurate channel — on your most important attribute.
- 5 Check the mark type hasn't already spent size or shape before you reach for them.
- 6 Keep channel counts below what people can discriminate; bin deliberately if not.
- 7 If two attributes must be read independently, pick a separable channel pair.
- 8 Give the eye a common frame — aligned baselines and gridlines beat bare comparison.

Apply these to the visualization you brought tonight. That is Assignment 1.

The whole Why? framework, on one page

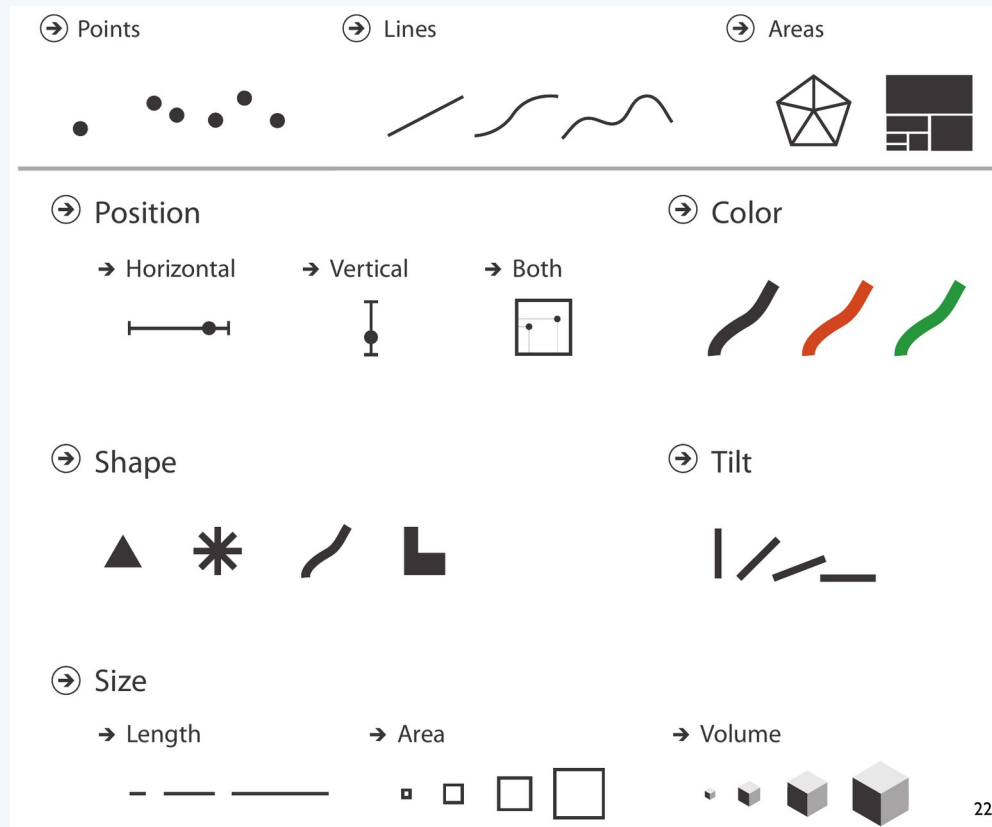
Worth printing. Actions on the left, targets on the right — this is the page to keep open while you do Assignment 1.



Munzner, Visualization Analysis and Design, Ch. 3 — figures via the SPU Information Visualization course deck (M. Senturk), after Brehmer & Munzner, IEEE TVCG 19(12), 2013.

And the whole How? vocabulary

Marks across the top, channels below. Between these two pages you can describe any static visualization.



Munzner, *Visualization Analysis and Design*, Ch. 5 — via the SPU Information Visualization course deck (M. Senturk), after Alex Lex & Miriah Meyer, dataviscourse.net.

Assignment 1 — and what to bring to Week 3

1 Read VAD Chapters 3 and 5

"Why: Task Abstraction" and "Marks and Channels."

These two chapters are the densest of the course and they carry heavy weight on the midterm.

2 Complete Assignment 1

Posted on Blackboard with the full prompt and due date.

You'll analyze published visualizations using tonight's vocabulary — task abstraction, then marks and channels.

3 Try the genomics exercise

The task-abstraction exercise from the reading, on gene expression across cancer samples.

Not graded. It is the best practice available for the midterm's abstraction questions.

4 Python must be working

matplotlib, pandas and seaborn. Week 3 begins with code.

If your environment is broken, post on the Blackboard discussion board this week, not next Monday.

5 Keep your two-number sketches

Photograph tonight's 9-and-26 sketches.

We will revisit them when we get to design alternatives and validation.

6 Tableau licence

If you still haven't requested it, do it this week.

Week 5 is the deadline and approval takes several days.

Next week: What — data abstraction — plus your first visualizations in Python.

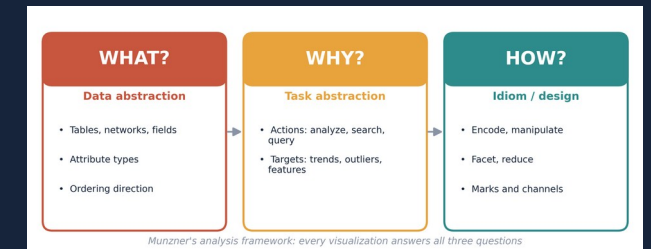
WEEK 2 COMPLETE

There are no good charts. Only good matches.

A design is good relative to a task, on a particular data abstraction, judged by how accurately its channels convey what matters. You can now say all three.

QUESTIONS

Reda MASTOURI · rmastouri@saintpeters.edu · 1 (888) 772-9933
Office hours and the discussion board are on Blackboard.



Two of the three answered. WHAT — data abstraction — is next week.

Sources and image credits

PRIMARY TEXT

Munzner, T. Visualization Analysis and Design. A K Peters/CRC Press. ISBN 9781466508910. Companion site: cs.ubc.ca/~tmm/vadbook/ Chapter 3, "Why: Task Abstraction", and Chapter 5, "Marks and Channels", are tonight's reading and the source of this deck's structure.

COURSE MATERIALS

Information Visualization course decks — Task Abstraction and Marks & Channels — by Metin Senturk, MS in Data Science, Saint Peter's University (github.com/spu-dataviz-211), which themselves credit Munzner VAD and Alex Lex & Miriah Meyer, dataviscourse.net. Task/chart rating exercise after Marti Hearst.

RESEARCH CITED

Brehmer & Munzner, A Multi-Level Typology of Abstract Visualization Tasks, IEEE TVCG 19(12):2376–2385, 2013. • Munzner, A Nested Model of Visualization Design and Validation, IEEE TVCG 15(6), 2009. • Cleveland & McGill, Graphical Perception, JASA 79:387 (1984), 531–554. • Heer & Bostock, Crowdsourcing Graphical Perception, CHI 2010, 203–212. • Auber, Using Strahler numbers for real-time visual exploration of huge graphs, 2002. • Riche & Dwyer, Untangling Euler Diagrams, 2010. • Strobel et al., Vials, 2016.

EXAMPLES & FIGURES

Horried? by Mohamad A. Waked (alhadaqa.com). • The Economist, Greece's GDP. • NYT Upshot, How the Recession Reshaped the Economy in 255 Charts. • Stephen Few, radar graph critique (perceptualedge.com). • Adelson checkershadow illusion (persci.mit.edu). • Purves Lab colour constancy (purveslab.net). • TeleGeography map (mappa.mundi.net). • Snowfall graphic via twitter.com/ChaseThomason.

ORIGINAL FIGURES

The action/target grammar, the search matrix, the query panels, the derive demonstration, the expressiveness/effectiveness panels, Steven's power law, the popout, separability, discriminability, redundant-encoding and Weber's-law demonstrations, the chart decomposition, the channel families and the course roadmap were generated for this deck.